



L4b: Single Asset Geometric Brownian Motion Models

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Objectives for Today

Today we introduce our first continuous-time price model, geometric Brownian motion, estimate its parameters from data, test it against the stylized facts, and compute a closed-form target probability for a scheduled trade.

Today's concepts

- **Model share prices with GBM**: the stochastic differential equation, the Wiener process, the exact lognormal solution, and the exact one-step transition used for Monte Carlo.
- **Estimate drift and volatility**: derive the growth-rate law, estimate the mean growth rate and volatility from a growth-rate series, and recover the arithmetic drift.
- **Test the model and evaluate a trade**: check GBM against the stylized facts, then derive the probability that a scheduled long position clears a target return.

Lectures and Examples

Lecture: Single-Asset Geometric Brownian Motion

Example: Estimating GBM Parameters

Estimate mean growth and volatility from historical prices, recover the arithmetic drift, and compare fitted paths with the data.

Example: GBM Terminal-Target Probabilities

Compute the closed-form probability that a scheduled long position clears a target present-value return.

Company Profile: Jane Street

Jane Street is a privately held **quantitative trading firm**: it trades across asset classes on more than 200 venues, deploys mainly its own capital, and is a major market maker in exchange-traded funds.

- **Client posture**: Citadel Securities handles a large share of U.S. retail order flow; Jane Street is primarily proprietary and institution facing.
- **Product emphasis**: Jane Street is synonymous with ETFs; Citadel Securities with U.S. equities, options, and FX market making.
- **Engineering culture**: Jane Street optimizes for correctness and composability (functional programming in OCaml); Citadel Securities for reliability and throughput at scale.

Role in this lecture: both firms depend on quantitative models of prices and execution; today we build one baseline price model.

Concept Review: N-Ary Tree Models

A binomial lattice allows only two price movements per step. If two possible futures are not enough, an **N-ary lattice** allows m branches with factors $f_1, \dots, f_m$ and probabilities $p_1, \dots, p_m$.

- A recombining node is indexed by its branch-count vector; its probability is multinomial (L4a).
- More branches make the set of possible prices finer (at the cost of more states); a smaller time step brings the steps closer together.

Today's idea: replace the discrete set of prices with a **continuous** one and the discrete steps with continuous time. Refinement alone is not enough; with the right scaling the lattice converges to a continuous model, and that model is geometric Brownian motion.

Example: [N-Ary Recombining Lattice Models](#)

Single Asset Geometric Brownian Motion (GBM)

Let $S(t) > 0$ be the share price at time t on a horizon $0 \leq t \leq T$. Let $\mu \in \mathbb{R}$ (units: inverse years) be a constant arithmetic drift, $\sigma > 0$ (units: inverse years to the one-half power) a constant **volatility**, and $dW(t)$ the increment of a Wiener process (next frame). Geometric Brownian motion (GBM) is a continuous-time stochastic process whose drift and noise are both proportional to the price:

$$\frac{dS(t)}{S(t)} = \underbrace{\mu dt}_{\text{drift}} + \underbrace{\sigma dW(t)}_{\text{noise}}.$$

We will see that the log price is a Brownian motion with drift. We reserve g for an observed continuously compounded growth rate. The drift μ and a growth rate g share inverse-time units, but they are different quantities; the relationship is made precise below.

The Wiener Process

A Wiener process $\{W(t), 0 \leq t \leq T\}$ is a continuous real-valued stochastic process with three properties:

- The noise vanishes at zero: $W(0) = 0$ with probability one.
- Increments over disjoint intervals $W(t_1) - W(t_0), \dots, W(t_k) - W(t_{k-1})$ are independent for any $0 \leq t_0 < t_1 < \dots < t_k \leq T$.
- Each increment is normal with mean zero and variance equal to the elapsed time, $W(t) - W(s) \sim \mathcal{N}(0, t - s)$ for $0 \leq s < t \leq T$.

The third property lets us write any increment as a scaled standard normal:

$$W(t) - W(s) = \sqrt{t - s} Z, \quad Z \sim \mathcal{N}(0, 1).$$

The Analytical Solution

Take $t_0 = 0$ and let S_0 be today's price. Solving the SDE with Itô calculus gives the share price at a fixed future time T :

$$S_T = S_0 \exp \left[\underbrace{\left(\mu - \frac{\sigma^2}{2} \right) T}_{\text{mean log growth}} + \underbrace{\sigma \sqrt{T} Z}_{\text{shock}} \right], \quad Z \sim \mathcal{N}(0, 1).$$

Thus the log price ratio $\ln(S_T/S_0)$ is normal with mean $(\mu - \sigma^2/2)T$ and variance $\sigma^2 T$, and the price S_T is **lognormal** and strictly positive.

The $-\sigma^2/2$ term comes out of the Itô calculus in the derivation notebook; it is the **half-variance correction** we meet again below.

Derivation: [Deriving the GBM Solution](#)

Mean and Variance of the GBM Price

From the lognormal moments, the expectation and variance of S_T are:

$$\underbrace{\mathbb{E}(S_T)}_{\text{mean price}} = S_0 e^{\mu T}, \quad \underbrace{\text{Var}(S_T)}_{\text{price variance}} = S_0^2 e^{2\mu T} (e^{\sigma^2 T} - 1).$$

The expected price grows at the arithmetic drift μ . The median price $S_0 e^{(\mu - \sigma^2/2)T}$ grows at the smaller rate $\mu - \sigma^2/2$.

The gap is the **half-variance correction**: the lognormal distribution is right skewed, so its mean sits above its median.

Discrete Time GBM Model

We observe prices on a grid $t_j = j\Delta t$, $j = 0, 1, \dots, N$, with time step Δt (years) and horizon $T = N\Delta t$. Applying the fixed-horizon solution over one step gives the **one-step transition**:

$$S_{t_j} = S_{t_{j-1}} \exp\left[\left(\mu - \frac{\sigma^2}{2}\right)\Delta t + \sigma\sqrt{\Delta t} Z_j\right], \quad j = 1, \dots, N,$$

with $Z_1, \dots, Z_N$ **independent** standard normals. For constant μ and σ this is exact at the grid points, not a numerical time-stepping approximation.

Jumping directly from S_0 to t_j uses the cumulative shock $\tilde{Z}_j = j^{-1/2} \sum_{i \leq j} Z_i$:

$$S_{t_j} = S_0 \exp\left[\left(\mu - \frac{\sigma^2}{2}\right)t_j + \sigma\sqrt{t_j} \tilde{Z}_j\right].$$

$\tilde{Z}_j$ is standard normal at each j but **not** independent across j (correlation $\sqrt{t_j/t_k}$ for $j < k$): use it for one horizon, use the transition for paths.

Monte Carlo Simulation of GBM

Given S_0 , μ , σ , Δt , N steps, and M paths, store the simulated prices in $\mathbf{S} \in \mathbb{R}^{(N+1) \times M}$, one path per column, rows indexed by $j = 0, \dots, N$.

For each sample path (each column of $\mathbf{S}$):

1. Set the price in row $j = 0$ to S_0 .
2. For $j = 1, \dots, N$: draw $Z_j \sim \mathcal{N}(0, 1)$ and apply the one-step transition:

$$S_{t_j} \leftarrow S_{t_{j-1}} \exp\left[\left(\mu - \frac{\sigma^2}{2}\right)\Delta t + \sigma\sqrt{\Delta t} Z_j\right].$$

The result is M possible future trajectories of N steps each. Their spread is process uncertainty; the uncertainty in μ and σ is a separate layer.

Implementation: the `sample(...)` function in the course package.

Estimation of GBM Parameters

Divide the one-step transition by $S_{t_{j-1}}$, take logarithms, and divide by Δt . The **one-step growth rate** g_j over $t_{j-1} \rightarrow t_j$, the same quantity we computed from data in L3a, is:

$$g_j \equiv \frac{1}{\Delta t} \ln \frac{S_{t_j}}{S_{t_{j-1}}} = \underbrace{\left(\mu - \frac{\sigma^2}{2}\right)}_{\text{mean growth } \bar{g}} + \underbrace{\frac{\sigma}{\sqrt{\Delta t}}}_{\text{growth-rate std}} Z_j.$$

Because the Z_j are independent standard normals, growth rates over equally spaced, non-overlapping steps are independent and normal with mean $\bar{g} \equiv \mu - \sigma^2/2$ and variance $\sigma^2/\Delta t$.

- **Drift versus growth:** μ is the SDE drift; the mean growth rate $\bar{g}$ is what a growth-rate series measures. They differ by the half-variance correction.
- **Brownian dispersion:** σ sets the growth-rate standard deviation $\sigma/\sqrt{\Delta t}$ and the one-step log-return standard deviation $\sigma\sqrt{\Delta t}$.

Volatility

From $N + 1$ prices on the grid we form N growth rates $g_1, \dots, g_N$. Their sample mean g' and sample standard deviation σ_g (our L3a risk measure) are:

$$g' = \frac{1}{N} \sum_{j=1}^N g_j, \quad \underbrace{\sigma_g}_{\text{growth-rate std}} = \sqrt{\frac{1}{N-1} \sum_{j=1}^N (g_j - g')^2}.$$

(Denominator $N - 1$: N observations less the one mean estimated from them.) But σ_g is **not** the GBM volatility. Since $\text{Var}(g_j) = \sigma^2 / \Delta t$, σ_g estimates $\sigma / \sqrt{\Delta t}$, and the volatility estimate is:

$$\hat{\sigma} = \sigma_g \sqrt{\Delta t}.$$

For daily data $\Delta t = 1/252$: divide the daily growth-rate standard deviation by $\sqrt{252}$, or equivalently multiply the daily log-return standard deviation by $\sqrt{252}$.

Drift

Under GBM the arithmetic drift is $\mu = \bar{g} + \sigma^2/2$. Given an estimate $\hat{g}$ of the mean growth rate, the drift estimate is:

$$\underbrace{\hat{\mu}}_{\text{SDE drift}} = \underbrace{\hat{g}}_{\text{mean growth}} + \underbrace{\frac{1}{2}\hat{\sigma}^2}_{\text{correction}}.$$

Two common ways to estimate $\bar{g}$:

- **Method 1, average growth rate:** $\hat{g} = g'$, the sample mean, which is also the maximum likelihood estimate under GBM.
- **Method 2, linear regression:** regress log price on time; the slope estimates $\bar{g}$, but the residuals are a Brownian path, so the ordinary standard errors and residual checks do not apply.

Mean growth is estimated less precisely than volatility because its signal is small relative to growth-rate noise; both methods therefore carry substantial sampling error in the drift.

Linear Regression: The Model

Take the logarithm of the cumulative GBM solution at grid point t_j :

$$\ln S_{t_j} = \underbrace{\ln S_0}_{\text{intercept}} + \underbrace{\bar{g} t_j}_{\text{trend}} + \underbrace{\sigma W(t_j)}_{\text{error}}, \quad j = 0, \dots, N.$$

The noise has mean zero, so the **expected** log price is a line with slope $\bar{g}$: regressing log price on time recovers that line, and the Wiener term is the error term.

That error is a Brownian path: neighboring values are strongly correlated, its variance $\sigma^2 t_j$ grows along the sample, and the fitted residuals inherit both features. Least squares still gives a sensible slope, but textbook standard errors, confidence intervals, and residual-normality tests assume independent, identically distributed errors and are not calibrated here.

Example 1: the Anderson--Darling normality test on these residuals rejects: a warning about the test's assumptions, not a verdict on GBM.

Linear Regression: The Normal Equations

Stack the $N + 1$ log prices as $\hat{\mathbf{X}}\theta + \epsilon = \mathbf{y}$, where $\theta = (\ln S_0, \bar{g})^\top$ contains the free intercept and slope:

$$\hat{\mathbf{X}} = \begin{bmatrix} 1 & t_0 \\ 1 & t_1 \\ \vdots & \vdots \\ 1 & t_N \end{bmatrix}, \quad \mathbf{y} = \begin{bmatrix} \ln S_{t_0} \\ \ln S_{t_1} \\ \vdots \\ \ln S_{t_N} \end{bmatrix}.$$

For full column rank, the least-squares solution is:

$$\hat{\theta} = (\hat{\mathbf{X}}^\top \hat{\mathbf{X}})^{-1} \hat{\mathbf{X}}^\top \mathbf{y}.$$

In Julia, compute it with `A \ y` rather than forming the inverse; the second component estimates $\bar{g}$.

Example: [Estimating GBM Parameters](#)

Stylized Facts

Does GBM capture the empirical regularities of asset growth rates?

- **Heavy tails**: empirical growth rates have heavier tails than a normal distribution, so extreme moves are more likely than a Gaussian predicts.
- **Absence of autocorrelation**: if prices follow a random walk, growth rates over non-overlapping intervals are uncorrelated, and there is no predictability to exploit.
- **Volatility clustering**: large moves tend to follow large moves and small moves follow small moves, so volatility alternates between high and low regimes.

We test each against the growth-rate law we just derived.

Growth Over a Horizon Under GBM

Define the **horizon-average growth rate** over $0 \rightarrow T$ as $g_{T,0} \equiv (1/T) \ln(S_T/S_0)$, in the same to-from subscript order as the discount factor $\mathcal{D}_{T,0}$. Substituting the GBM solution and dividing by T gives:

$$\underbrace{g_{T,0}}_{\text{horizon-average growth}} = \bar{g} + \frac{\sigma}{\sqrt{T}} Z, \quad Z \sim \mathcal{N}(0, 1).$$

So $g_{T,0}$ is normal with mean $\bar{g}$ and variance σ^2/T ; the one-step case is $T = \Delta t$.

The uncertainty of the horizon-average growth rate shrinks as T grows while its mean stays put. The price does not settle down: $\ln(S_T/S_0) = T g_{T,0}$ has variance $\sigma^2 T$, which grows with T .

Heavy Tails and Volatility Clustering

Heavy tails: not captured. One-step and horizon-average growth rates are both normal under GBM. The lognormal price has a long right tail, but that is not a heavy-tailed **growth-rate** distribution.

Volatility clustering: not captured. The volatility σ is constant, so there is no mechanism for high- and low-volatility periods to alternate.

Both failures follow from the same assumption: independent Gaussian increments with a constant variance rate. That assumption is what makes the model exactly solvable.

Autocorrelation of One-Step Growth Rates

From the growth-rate law, $g_j = \bar{g} + (\sigma/\sqrt{\Delta t})Z_j$ with independent shocks. The autocovariance at lag $\tau \geq 1$ is:

$$\text{Cov}(g_j, g_{j+\tau}) = \frac{\sigma^2}{\Delta t} \text{Cov}(Z_j, Z_{j+\tau}) = 0,$$

because Z_j and $Z_{j+\tau}$ are independent. With $\text{Var}(g_j) = \sigma^2/\Delta t$, the autocorrelation function is:

$$\rho_g(\tau) = \frac{\text{Cov}(g_j, g_{j+\tau})}{\text{Var}(g_j)} = 0, \quad \tau \geq 1.$$

Absence of autocorrelation: captured for one-step growth rates over non-overlapping intervals. This says nothing about squared growth rates, whose dependence (L3a) GBM also cannot produce.

GBM Trade Rule

Buy $n_0 > 0$ shares at S_0 at time 0; sell all n_0 shares at S_T at the scheduled exit $T = N\Delta t$, $N \geq 1$ (frictionless: no costs, price return only). Following L1b and L4a, let g_y denote the continuously compounded annual growth rate associated with the selected benchmark yield y . Then $\mathcal{D}_{T,0}(g_y) = e^{g_y T}$ is the accumulation factor, and $\mathcal{D}_{T,0}^{-1}(g_y) = e^{-g_y T}$ discounts the sale to time 0:

$$\text{NPV}(g_y, T) = \underbrace{-n_0 S_0}_{\text{entry (now)}} + \underbrace{n_0 S_T \mathcal{D}_{T,0}^{-1}(g_y)}_{\text{exit (future)}}.$$

Dividing by the initial investment gives the **scaled NPV**, a dimensionless present-value return on the share cost:

$$\underbrace{\rho_T}_{\text{scaled NPV}} := \frac{\text{NPV}(g_y, T)}{n_0 S_0} = \left(\frac{S_T}{S_0} \right) e^{-g_y T} - 1.$$

This is the L4a rule with one change: S_T now follows GBM instead of a binomial lattice.

Substitute the GBM Solution

Replacing S_T by the GBM solution puts all of the randomness in one standard normal Z :

$$\rho_T = \exp\left[\left(\mu - \frac{\sigma^2}{2}\right)T + \sigma\sqrt{T} Z\right] \underbrace{e^{-g_y T}}_{\text{discount}} - 1, \quad Z \sim \mathcal{N}(0, 1).$$

- **Short holding period:** for a few days or weeks and a realistic benchmark, $|g_y|T$ is small, $e^{-g_y T} \approx 1$, and $\rho_T \approx S_T/S_0 - 1$ is the fractional price change.
- **Longer holding period:** whenever $|g_y|T$ is not small, keep the discount factor.

Either way ρ_T is a continuous random variable driven by Z , so a target probability is a one-line standardization.

Cumulative Probability of a Trade Exit

Let $\rho_\star > -1$ be a target (under GBM $S_T > 0$, so $\rho_T > -1$ always). We ask for the probability that ρ_T at the scheduled exit **strictly exceeds** the target. Moving the non-random terms to the right:

$$\mathbb{P}(\rho_T > \rho_\star) = \mathbb{P}\left(\ln \frac{S_T}{S_0} > \ln(1 + \rho_\star) + g_y T\right).$$

Under GBM, $\ln(S_T/S_0)$ is normal with mean $(\mu - \sigma^2/2)T$ and standard deviation $\sigma\sqrt{T}$. Standardizing both sides:

$$\mathbb{P}(\rho_T > \rho_\star) = \mathbb{P}\left(\underbrace{\frac{\ln(S_T/S_0) - (\mu - \sigma^2/2)T}{\sigma\sqrt{T}}}_{Z \sim \mathcal{N}(0,1)} > \frac{\ln(1 + \rho_\star) + g_y T - (\mu - \sigma^2/2)T}{\sigma\sqrt{T}}\right).$$

As in L4a, this is the scheduled terminal exit; a monitored take-profit or stop-loss is a first-passage question.

The Closed-Form Target Probability

A standard normal exceeds z with probability $1 - \Phi(z)$, where Φ is the standard normal cumulative distribution function. Therefore:

$$\mathbb{P}(\rho_T > \rho_\star) = 1 - \Phi\left(\frac{\ln(1 + \rho_\star) + g_y T - (\mu - \frac{\sigma^2}{2})T}{\sigma\sqrt{T}}\right).$$

- The probability uses the **real-world** μ and σ estimated from data. A risk-neutral drift, which we will use to price derivatives later, changes the measure and no longer answers a probability-of-profit question.
- For a short holding period $g_y T \approx 0$ and the benchmark drops out of the numerator.
- A target of -1 or below is cleared with probability one.

Example: [GBM Terminal-Target Probabilities](#)

Optional Advanced Material

Advanced: [From the Binomial Lattice to GBM](#)

Calibrate the lattice to GBM and track convergence of its terminal distribution and target probability.

Advanced: [First-Passage Rules Under GBM](#)

Compare daily and continuous monitoring for single- and two-barrier exit rules.

Advanced: [Drift Uncertainty](#)

Derive the calendar-span rule and propagate drift uncertainty into a target probability.

Advanced: [Monte Carlo Target Probabilities](#)

Add sampling error bars and compare exact GBM simulation with Euler time stepping.

Optional; none of these notebooks is a prerequisite for L5a.

Summary

Today we introduced geometric Brownian motion as our first continuous-time price model, estimated its parameters from growth-rate data, tested it against the stylized facts, and derived a closed-form probability for a scheduled trade.

Key takeaways

- **GBM is a tractable lognormal model:** exact solution, exact one-step simulation with independent normal shocks, mean growing at the drift.
- **Drift and growth are different parameters:** the growth-rate series gives $\bar{g}$ and, through $\sigma_g \sqrt{\Delta t}$, the volatility; the drift follows from the half-variance correction.
- **One stylized fact passed, two failed:** growth rates are uncorrelated but Gaussian with constant volatility; the terminal target probability is one minus a standard normal CDF.

Next, we extend GBM to several correlated assets and use it to build portfolios.

Today: the lattice becomes a continuous process, and the trade rule gets a closed form.